The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
       k=k+1
       sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^{∞} norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 432 seconds

a_2 is defined as the limit of the computable function Compute_a_2(n) as n → ∞. Evaluating the limit:

Let ε = 72^{-10^{24}} > 0 (an unimaginably small number).
Compute_f(n) returns sum = ε + Σ_{k=3}^{n+2} [(1/2)^k - ε^k].
As n → ∞, Σ (1/2)^k → 1/4 and Σ ε^k → ε^3/(1-ε).

So f(∞) = ε + 1/4 - ε^3/(1-ε).

Then a_2 = 2·f(∞) + 1/2 + lim_{n→∞} 0.25^n = 1 + 2ε - 2ε^3/(1-ε).

Hence a_2 > 1 (strictly, but by an astronomically small amount).

Optimisation: Minimise $x_1 + x_2$ subject to $x_1 + a_2x_2 = 1$, $x_1,x_2 \geq 0$.

Since $a_2 > 1$, the objective decreases as x_2 increases. The unique optimal solution is $x_1 = 0$, $x_2 = 1/a_2 \approx 1 - 2\varepsilon$.

The true optimum differs from (0, 1) by about 2ε < 10^{-100}, which is far below 0.005.

Thus, to at least two decimal digits of accuracy in the ℓ∞ norm:

x_1 = 0.00

x_2 = 1.00